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Grafana

Grafana Alloy

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INSTALL

Grafana Alloy Helm chart

Type application Version 1.0.1 AppVersion v1.8.1

Helm chart for deploying Grafana Alloy to Kubernetes.

Usage

Setup Grafana chart repository

```
helm repo add grafana https://grafana.github.io/helm-charts
helm repo update
```

Install chart

To install the chart with the release name my-release:

`helm install my-release grafana/alloy`

This chart installs one instance of Grafana Alloy into your Kubernetes cluster using a specific Kubernetes controller. By default, DaemonSet is used. The `controller.type` value can be used to change the controller to either a StatefulSet or Deployment.

Creating multiple installations of the Helm chart with different controllers is useful if just using the default DaemonSet isn't sufficient.

Values

Key	Type	Default
alloy.clustering.enabled	bool	`false`
alloy.clustering.name	string	`""`
alloy.clustering.portName	string	`"http"`
alloy.configMap.content	string	`""`
alloy.configMap.create	bool	`true`
alloy.configMap.key	string	`nil`
alloy.configMap.name	string	`nil`
alloy.enableReporting	bool	`true`
alloy.envFrom	list	`[]`
alloy.extraArgs	list	`[]`
alloy.extraEnv	list	`[]`
alloy.extraPorts	list	`[]`
alloy.hostAliases	list	`[]`
alloy.lifecycle	object	`{}`
alloy.listenAddr	string	`"0.0.0.0"`
alloy.listenPort	int	`12345`
alloy.listenScheme	string	`"HTTP"`
alloy.livenessProbe	object	`{}`
alloy.mounts.dockercontainers	bool	`false`
alloy.mounts.extra	list	`[]`
alloy.mounts.varlog	bool	`false`

Key	Type	Default
alloy.resources	object	`{}`
alloy.securityContext	object	`{}`
alloy.stabilityLevel	string	`"generally-available"`
alloy.storagePath	string	`"/tmp/alloy"`
alloy.uiPathPrefix	string	`"/"`
configReloader.customArgs	list	`[]`
configReloader.enabled	bool	`true`
configReloader.image.digest	string	`""`
configReloader.image.registry	string	`"quay.io"`
configReloader.image.repository	string	`"prometheus-operator/prometheus-config-reloader"`
configReloader.image.tag	string	`"v0.81.0"`
configReloader.resources	object	`{"requests":{"cpu":"10m","memory":"50Mi"}}`
configReloader.securityContext	object	`{}`
controller.affinity	object	`{}`
controller.autoscaling.enabled	bool	`false`
controller.autoscaling.horizontal	object	`{"enabled":false,"maxReplicas":5,"minReplicas":1,"scaleDown":{"policies":[],"selectPolicy":"Max","stabilizationWindowSeconds":300},"scaleUp":{"policies":[],"selectPolicy":"Max","stabilizationWindowSeconds":0},"targetCPUUtilizationPercentage":0,"targetMemoryUtilizationPercentage":80}`
controller.autoscaling.horizontal.enabled	bool	`false`
controller.autoscaling.horizontal.maxReplicas	int	`5`
controller.autoscaling.horizontal.minReplicas	int	`1`
controller.autoscaling.horizontal.scaleDown.policies	list	`[]`
controller.autoscaling.horizontal.scaleDown.selectPolicy	string	`"Max"`
controller.autoscaling.horizontal.scaleDown.stabilizationWindowSeconds	int	`300`
controller.autoscaling.horizontal.scaleUp.policies	list	`[]`
controller.autoscaling.horizontal.scaleUp.selectPolicy	string	`"Max"`
controller.autoscaling.horizontal.scaleUp.stabilizationWindowSeconds	int	`0`
controller.autoscaling.horizontal.targetCPUUtilizationPercentage	int	`0`
controller.autoscaling.horizontal.targetMemoryUtilizationPercentage	int	`80`
controller.autoscaling.maxReplicas	int	`5`
controller.autoscaling.minReplicas	int	`1`
controller.autoscaling.scaleDown.policies	list	`[]`
controller.autoscaling.scaleDown.selectPolicy	string	`"Max"`
controller.autoscaling.scaleDown.stabilizationWindowSeconds	int	`300`
controller.autoscaling.scaleUp.policies	list	`[]`
controller.autoscaling.scaleUp.selectPolicy	string	`"Max"`
controller.autoscaling.scaleUp.stabilizationWindowSeconds	int	`0`
controller.autoscaling.targetCPUUtilizationPercentage	int	`0`
controller.autoscaling.targetMemoryUtilizationPercentage	int	`80`
controller.autoscaling.vertical	object	`{"enabled":false,"recommenders":[],"resourcePolicy":{"containerPolicies":[{"containerName":"alloy","controlledResources":["cpu","memory"],"controlledValues":"RequestsAndLimits","maxAllowed":{},"minAllowed":{}}]},"updatePolicy":null}`
controller.autoscaling.vertical.enabled	bool	`false`
controller.autoscaling.vertical.recommenders	list	`[]`

Key	Type	Default
controller.autoscaling.vertical.resourcePolicy	object	`{"containerPolicies":[{"containerName":"alloy","controlledResources":["cpu","memory"],"controlledValues":"RequestsAndLimits","maxAllowed":{},"minAllowed":{}}]}`
controller.autoscaling.vertical.resourcePolicy.containerPolicies	list	`[{"containerName":"alloy","controlledResources":["cpu","memory"],"controlledValues":"RequestsAndLimits","maxAllowed":{},"minAllowed":{}}]`
controller.autoscaling.vertical.resourcePolicy.containerPolicies[0].controlledResources	list	`["cpu","memory"]`
controller.autoscaling.vertical.resourcePolicy.containerPolicies[0].controlledValues	string	`"RequestsAndLimits"`
controller.autoscaling.vertical.resourcePolicy.containerPolicies[0].maxAllowed	object	`{}`
controller.autoscaling.vertical.resourcePolicy.containerPolicies[0].minAllowed	object	`{}`
controller.autoscaling.vertical.updatePolicy	string	`nil`
controller.dnsPolicy	string	`"ClusterFirst"`
controller.enableStatefulSetAutoDeletePVC	bool	`false`
controller.extraAnnotations	object	`{}`
controller.extraContainers	list	`[]`
controller.hostNetwork	bool	`false`
controller.hostPID	bool	`false`
controller.initContainers	list	`[]`
controller.nodeSelector	object	`{}`
controller.parallelRollout	bool	`true`
controller.podAnnotations	object	`{}`
controller.podDisruptionBudget	object	`{"enabled":false,"maxUnavailable":null,"minAvailable":null}`
controller.podDisruptionBudget.enabled	bool	`false`
controller.podDisruptionBudget.maxUnavailable	string	`nil`
controller.podDisruptionBudget.minAvailable	string	`nil`
controller.podLabels	object	`{}`
controller.priorityClassName	string	`""`
controller.replicas	int	`1`
controller.terminationGracePeriodSeconds	string	`nil`
controller.tolerations	list	`[]`
controller.topologySpreadConstraints	list	`[]`
controller.type	string	`"daemonset"`
controller.updateStrategy	object	`{}`
controller.volumeClaimTemplates	list	`[]`
controller.volumes.extra	list	`[]`
crds.create	bool	`true`
extraObjects	list	`[]`
fullnameOverride	string	`nil`
global.image.pullSecrets	list	`[]`
global.image.registry	string	`""`
global.podSecurityContext	object	`{}`
image.digest	string	`nil`
image.pullPolicy	string	`"IfNotPresent"`
image.pullSecrets	list	`[]`
image.registry	string	`"docker.io"`
image.repository	string	`"grafana/alloy"`
image.tag	string	`nil`
ingress.annotations	object	`{}`
ingress.enabled	bool	`false`
ingress.extraPaths	list	`[]`
ingress.faroPort	int	`12347`
ingress.hosts[0]	string	`"chart-example.local"`
ingress.labels	object	`{}`
ingress.path	string	`"/"`

Key	Type	Default
ingress.pathType	string	"Prefix"
ingress.tls	list	[]
nameOverride	string	nil
namespaceOverride	string	nil
rbac.create	bool	true
service.annotations	object	{}
service.clusterIP	string	""
service.enabled	bool	true
service.internalTrafficPolicy	string	"Cluster"
service.nodePort	int	31128
service.type	string	"ClusterIP"
serviceAccount.additionalLabels	object	{}
serviceAccount.annotations	object	{}
serviceAccount.automountServiceAccountToken	bool	true
serviceAccount.create	bool	true
serviceAccount.name	string	nil
serviceMonitor.additionalLabels	object	{}
serviceMonitor.enabled	bool	false
serviceMonitor.interval	string	""
serviceMonitor.metricRelabelings	list	[]
serviceMonitor.relabelings	list	[]
serviceMonitor.tlsConfig	object	{}

Migrate from `grafana/grafana-agent` chart to `grafana/alloy`

The `values.yaml` file for the `grafana/grafana-agent` chart is compatible with the chart for `grafana/alloy`, with two exceptions:

- The `agent` field in `values.yaml` is deprecated in favor of `alloy`. Support for the `agent` field will be removed in a future release.
- The default value for `alloy.listenPort` is `12345` to align with the default listen port in other installations. To retain the previous default, set `alloy.listenPort` to `80` when installing.

alloy.stabilityLevel

`alloy.stabilityLevel` controls the minimum level of stability for what components can be created (directly or through imported modules). Note that setting this field to a lower stability may also enable internal behaviour of a lower stability, such as experimental memory optimizations.

Valid settings are `experimental`, `public-preview`, and `generally-available`.

alloy.extraArgs

`alloy.extraArgs` allows for passing extra arguments to the Grafana Alloy container. The list of available arguments is documented on alloy run.

> **WARNING**: Using `alloy.extraArgs` does not have a stable API. Things may break between Chart upgrade if an argument gets added to the template.

alloy.extraPorts

`alloy.extraPorts` allows for configuring specific open ports.

The detained specification of ports can be found at the Kubernetes Pod documents.

Port numbers specified must be 0 < x < 65535.

ChartPort	KubePort	Description
targetPort	containerPort	Number of port to expose on the pod's IP address.
hostPort	hostPort	(Optional) Number of port to expose on the host. Daemonsets taking traffic might find this useful.
name	name	If specified, this must be an `IANA_SVC_NAME` and unique within the pod. Each named port in a pod must have a unique name. Name for the port that can be referred to by services.
protocol	protocol	Must be UDP, TCP, or SCTP. Defaults to "TCP".
appProtocol	appProtocol	Hint on application protocol. This is used to expose Alloy externally on OpenShift clusters using "h2c". Optional. No default value.

alloy.listenAddr

`alloy.listenAddr` allows for restricting which address Alloy listens on for network traffic on its HTTP server. By default, this is `0.0.0.0` to allow its UI to be exposed when port-forwarding and to expose its metrics to other Alloy instances in the cluster.

alloy.configMap.config

`alloy.configMap.content` holds the Grafana Alloy configuration to use.

If `alloy.configMap.content` is not provided, a default configuration file is used. When provided, `alloy.configMap.content` must hold a valid Alloy configuration file.

alloy.securityContext

`alloy.securityContext` sets the securityContext passed to the Grafana Alloy container.

By default, Grafana Alloy containers are not able to collect telemetry from the host node or other specific types of privileged telemetry data. See [Collecting logs from other containers][#collecting-logs-from-other-containers] and [Collecting host node telemetry][#collecting-host-node-telemetry] below for more information on how to enable these capabilities.

rbac.create

`rbac.create` enables the creation of ClusterRole and ClusterRoleBindings for the Grafana Alloy containers to use. The default permission set allows components like discovery.kubernetes to work properly.

controller.autoscaling

`controller.autoscaling.enabled` enables the creation of a HorizontalPodAutoscaler. It is only used when `controller.type` is set to `deployment` or `statefulset`.

`controller.autoscaling` is intended to be used with clustered mode.

> **WARNING**: Using `controller.autoscaling` for any other Grafana Alloy configuration could lead to redundant or double telemetry collection.

When using autoscaling with a StatefulSet controller and have enabled volumeClaimTemplates to be created alongside the StatefulSet, it is possible to leak up to `maxReplicas` PVCs when the HPA is scaling down. If you're on Kubernetes version `>=1.23-0` and your cluster has the `StatefulSetAutoDeletePVC` feature gate enabled, you can set `enableStatefulSetAutoDeletePVC` to true to automatically delete stale PVCs.

Using `controller.autoscaling` requires the target metric (cpu/memory) to have its resource requests set up for both the Alloy and config-reloader containers so that the HPA can use them to calculate the replica count from the actual resource utilization.

Collecting logs from other containers

There are two ways to collect logs from other containers within the cluster Alloy is deployed in.

loki.source.kubernetes

The loki.source.kubernetes component may be used to collect logs from containers using the Kubernetes API. This component does not require mounting the hosts filesystem into Alloy, nor requires additional security contexts to work correctly.

File-based collection

Logs may also be collected by mounting the host's filesystem into the Alloy container, bypassing the need to communicate with the Kubrnetes API.

To mount logs from other containers to Grafana Alloy directly:

- Set `alloy.mounts.dockercontainers` to `true`.
- Set `alloy.securityContext` to:

  ```
  privileged: true
  runAsUser: 0
  ```

Collecting host node telemetry

Telemetry from the host, such as host-specific log files (from `/var/logs`) or metrics from `/proc` and `/sys` are not accessible to Grafana Alloy containers.

To expose this information to Grafana Alloy for telemetry collection:

- Set `alloy.mounts.dockercontainers` to `true`.
- Mount `/proc` and `/sys` from the host into the container.
- Set `alloy.securityContext` to:

  ```
  privileged: true
  runAsUser: 0
  ```

Expose Alloy externally on OpenShift clusters

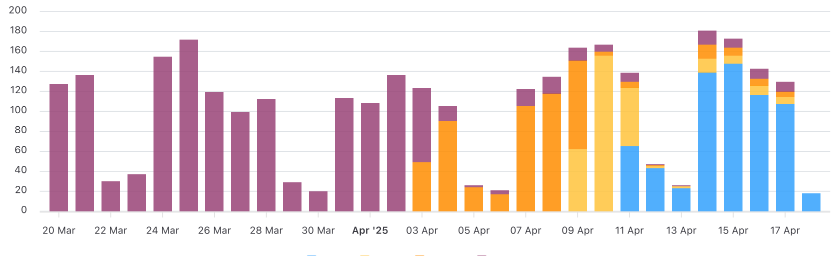
If you want to send telemetry from an Alloy instance outside of the OpenShift clusters over gRPC towards the Alloy instance on the OpenShift clusters, you need to:

- Set the optional `appProtocol` on `alloy.extraPorts` to `h2c`
- Expose the service via Ingress or Route within the OpenShift cluster. Example of a Route in OpenShift:

```
kind: Route
apiVersion: route.openshift.io/v1
metadata:
  name: route-otlp-alloy-h2c
spec:
  to:
    kind: Service
    name: test-grpc-h2c
    weight: 100
  port:
    targetPort: otlp-grpc
  tls:
    termination: edge
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: None
```

Once this Ingress/Route is exposed it would then allow gRPC communication for (for example) traces. This allow an Alloy instance on a VM or another Kubernetes/OpenShift cluster to be able to communicate over gRPC via the exposed Ingress or Route.

Views over the last 30 days



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